

## Answer Key: Lesson 01, Square and Cube Roots

Teacher document. Complete worked solutions. The After-Lesson Practice and Self-Test sections may be released to students after marking.

### Conventions used throughout

$\sqrt{\quad}$  means the positive (principal) square root. Estimates are to 1 decimal place (1 d.p.) using the four-step method: (1) square/cube numbers either side, (2) which is it closer to, (3) trap between decimals, (4) choose the decimal whose square/cube is closer, then check. For every estimate in this package, Step 4 agrees with the true rounded value (verified by calculator values given). **Accept** a correct answer by any valid method, including calculator checks where allowed.

### Preparation (Lesson\_01\_Preparation)

**P1.** (a)  $7 \times 7 = 49$  (b)  $12 \times 12 = 144$  (c)  $3 \times 3 \times 3 = 9 \times 3 = 27$

**P2.**  $5 \times 5 \times 5 = 5^3$ ;  $5 \times 5 = 25$ ,  $25 \times 5 = 125$ . So  $5^3 = 125$ .

**P3.** No.  $5^2 = 5 \times 5 = 25$ . Omar worked out  $5 \times 2$ ; the small 2 means “multiply 5 by itself”, not “multiply by 2”.

**P4.** Area =  $6^2 = 6 \times 6 = 36 \text{ cm}^2$  (units required:  $\text{cm}^2$ ).

**T1.**

| $n$ | $n^2$ | $n^3$ |
|-----|-------|-------|
| 3   | 9     | 27    |
| 4   | 16    | 64    |
| 5   | 25    | 125   |
| 6   | 36    | 216   |
| 10  | 100   | 1000  |

**T2.** (a)  $9^2 = 81$  (b)  $(-4)^2 = (-4) \times (-4) = 16$  (c)  $(-1)^3 = (-1) \times (-1) \times (-1) = 1 \times (-1) = -1$

**T3.**  $4^3 = 4 \times 4 \times 4 = 64$  and  $8^2 = 8 \times 8 = 64$ . They are **equal**. (Accept “neither, both are 64”.)

**N1.** (a) 7 cm, because  $7 \times 7 = 49$  (working backwards from the area).

(b) Not a whole number.  $49 < 50 < 64$ , so the side is between 7 and 8 cm, and very close to 7 (it is 7.07... cm). Any sensible notice/wonder is acceptable, e.g. “Is there a decimal that squares to exactly 50?”

### Two-minute plan for common Preparation errors (use at 04:00, slide 4)

$5^2 = 10$  or  $3^3 = 9$ : draw a  $5 \times 5$  dot square, count 25; say “the power tells you how many times to write the number, then multiply”.  $(-4)^2 = -16$ : “negative times negative is positive”; write  $(-4) \times (-4) = +16$ . **T3 “8<sup>2</sup> is bigger”**: compute both on the board. **N1(b) “25”** (halving): link to the misconception slide on halving. Students who did not do the Preparation answer N1(a) orally from the Starter.

### Starter (Class Worksheet page 1; slides 2–3)

**S1.**  $8^2 = 64$     **S2.**  $11^2 = 121$     **S3.**  $3^3 = 27$     **S4.**  $10^3 = 1000$     **S5.**  $4^3 = 64$

**S6.** Square numbers: 25 ( $5^2$ ), 36 ( $6^2$ ), 81 ( $9^2$ ). Not square: 20 (between 16 and 25), 48 (between 36 and 49).

## Mini-whiteboards (slides 11–12 and 16–17)

### Round 1: exact roots

**R1.1**  $\sqrt{9} = 3$  since  $3^2 = 9$ .      **R1.2**  $\sqrt{100} = 10$  since  $10^2 = 100$ .

**R1.3**  $\sqrt[3]{64} = 4$  since  $4^3 = 64$  (not 8:  $8^3 = 512$ ).

**R1.4** Side =  $\sqrt{81} = 9$  cm.

**R1.5 False.**  $\sqrt{36} = 6$  because  $6^2 = 36$ .  $18^2 = 324$ .

### Round 2: estimating

**R2.1**  $16 < 20 < 25$ , so  $4 < \sqrt{20} < 5$ .

**R2.2**  $36 < 45 < 49$ , so between 6 and 7.  $45 - 36 = 9$ ,  $49 - 45 = 4$ : closer to 7. ( $\sqrt{45} = 6.708\dots$ )

**R2.3** Cube numbers:  $27 < 30 < 64$ , so  $3 < \sqrt[3]{30} < 4$  (closer to 3;  $\sqrt[3]{30} = 3.107\dots$ ).

**R2.4**  $9 < 10 < 16$  so  $3 < \sqrt{10} < 4$ , closer to 3.  $3.1^2 = 9.61$ ,  $3.2^2 = 10.24$ . Differences:  $10 - 9.61 = 0.39$ ,  $10.24 - 10 = 0.24$ . So  $\sqrt{10} \approx 3.2$ . (Calculator 3.162...)

**R2.5 False.** 65 is only 1 more than 64, so  $\sqrt{65}$  is just above 8.  $8.0^2 = 64$ ,  $8.1^2 = 65.61$ : differences 1 and 0.61, so  $\sqrt{65} \approx 8.1$ . (Calculator 8.062...)

## I DO and WE DO (Class Worksheet pages 2–3; slides 6–15)

**I DO** (worked in full on the worksheet): Example 1:  $\sqrt{81} = 9$ ,  $\sqrt[3]{125} = 5$ ,  $\sqrt[3]{-27} = -3$ . Example 2:  $\sqrt{40} \approx 6.3$  ( $6.3^2 = 39.69$ ,  $6.4^2 = 40.96$ ; calculator 6.324...). Example 3:  $\sqrt[3]{50} \approx 3.7$  ( $3.6^3 = 46.656$ ,  $3.7^3 = 50.653$ ; calculator 3.684...).

**N1.** Square root undoes squaring. Sides:  $\sqrt{25} = 5$ ,  $\sqrt{36} = 6$ ,  $\sqrt{49} = 7$ . The square with area 40 has a side between 6 and 7 that is not a whole number (about 6.3).

**W1.** (a)  $12 \times 12 = 144$ , so  $\sqrt{144} = 12$ . Check  $12^2 = 144$ .      (b)  $10 \times 10 \times 10 = 1000$ , so  $\sqrt[3]{1000} = 10$ .

**W2.**  $64 < 70 < 81$ , so  $8 < \sqrt{70} < 9$ . Closer to 8 because  $70 - 64 = 6$  is smaller than  $81 - 70 = 11$ .  
Trap:  $8.3^2 = 68.89$  (too small),  $8.4^2 = 70.56$  (too big). Differences:  $70 - 68.89 = 1.11$ ,  $70.56 - 70 = 0.56$ .  
So  $\sqrt{70} \approx 8.4$ . (Calculator 8.366...) *Accept* other correct decimal pairs as long as the final pair is 8.3/8.4.

**W3.**  $8 < 20 < 27$ , so  $2 < \sqrt[3]{20} < 3$ .  $20 - 8 = 12$ ,  $27 - 20 = 7$ : closer to 3.  
 $2.7^3 = 19.683$ ,  $2.8^3 = 21.952$ . Differences 0.317 and 1.952. So  $\sqrt[3]{20} \approx 2.7$ . (Calculator 2.714...)

**W4.**  $\sqrt{1600} = 40$  since  $40^2 = 1600$ .  $\sqrt{0.16} = 0.4$  since  $0.4^2 = 0.16$ . *Alternative:*  $\sqrt{1600} = \sqrt{16} \times \sqrt{100} = 4 \times 10$ .

## Class Worksheet: YOU DO (pages 4–6)

### Foundation

**F1.** 5    **F2.** 8    **F3.** 11    **F4.** 14 ( $14^2 = 196$ )    **F5.** 2    **F6.** 4    **F7.** 10    **F8.** 1

### Core

**C1.**  $25 < 30 < 36$ , so  $5 < \sqrt{30} < 6$ .

**C2.**  $64 < 100 < 125$ , so  $4 < \sqrt[3]{100} < 5$ .

- C3.**  $49 < 55 < 64$ , so  $7 < \sqrt{55} < 8$ ;  $55 - 49 = 6$ ,  $64 - 55 = 9$ , closer to 7.  
 $7.4^2 = 54.76$ ,  $7.5^2 = 56.25$ . Differences 0.24 and 1.25.  $\sqrt{55} \approx 7.4$ . (Calculator 7.416... ✓)
- C4.**  $81 < 90 < 100$ , so  $9 < \sqrt{90} < 10$ , closer to 9 (9 vs 10 away:  $90 - 81 = 9$ ,  $100 - 90 = 10$ , only slightly).  
 $9.4^2 = 88.36$ ,  $9.5^2 = 90.25$ . Differences 1.64 and 0.25.  $\sqrt{90} \approx 9.5$ . (Calculator 9.486... ✓)  
*Teacher note:* a good example of why “closer to 9” does not mean the decimal is small.
- C5.**  $\sqrt[3]{-64} = -4$ . Check:  $(-4)^3 = (-4)(-4)(-4) = 16 \times (-4) = -64$ .
- C6.**  $\sqrt{45}$  is between 6 and 7 ( $36 < 45 < 49$ , about 6.7);  $\sqrt[3]{343} = 7$ ;  $\sqrt{50}$  is just above 7 ( $49 < 50$ , about 7.07); 7.5.  
 Order:  $\sqrt{45}$ ,  $\sqrt[3]{343}$ ,  $\sqrt{50}$ , 7.5. Check  $\sqrt{50} < 7.5$  because  $7.5^2 = 56.25 > 50$ .

### Apply

- A1.** Side =  $\sqrt{200}$ .  $196 < 200 < 225$ , so  $14 < \sqrt{200} < 15$ , closer to 14.  $14.1^2 = 198.81$ ,  $14.2^2 = 201.64$ . Differences 1.19 and 1.64. Side  $\approx 14.1$  m. (Calculator 14.142...)
- A2.** Edge =  $\sqrt[3]{729} = 9$  cm, since  $9^3 = 81 \times 9 = 729$ .
- A3.**  $\sqrt{900} = 30$  ( $30^2 = 900$ );  $\sqrt{0.09} = 0.3$  ( $0.3^2 = 0.09$ ). *Common error:* 0.03 ( $0.03^2 = 0.0009$ ).
- A4.** Any of  $n = 43, 44, 45, 46, 47, 48$ . Reason:  $\sqrt{n}$  is between 6 and 7 when  $36 < n < 49$ . It is closer to 7 when  $\sqrt{n} > 6.5$ , i.e.  $n > 6.5^2 = 42.25$ . *Accept* the explanation “ $n$  is closer to 49 than to 36” for  $n = 43-48$  (this gives the same set here).  $n = 42$  is **not** correct:  $\sqrt{42} = 6.48...$

### Challenge

- Ch1.** 729. Method: list cube numbers and test if each is square: 125 (no), 216 (no), 343 (no), 512 (no), 729 =  $27^2$  (yes). So  $729 = 27^2 = 9^3$ . (These numbers are sixth powers:  $1^6, 2^6 = 64, 3^6 = 729$ .)
- Ch2.** No.  $\sqrt{9 + 16} = \sqrt{25} = 5$ , but  $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$ .  $5 \neq 7$ . You must add inside the root first; roots do not “split” over addition.

### Justification (page 6)

- J1.**  $\sqrt[3]{20}$ :  $8 < 20 < 27$ , so between 2 and 3, closer to 3 (about 2.7).  
 $\sqrt{20}$ :  $16 < 20 < 25$ , so between 4 and 5, a little closer to 5 (about 4.5).  
 $\sqrt{50}$ :  $49 < 50 < 64$ , so just above 7 (about 7.1).  
 Accept marks placed in the correct interval on the correct side of the midpoint, with the bounding numbers written.

## Error Analysis: Spot the Mistake (pages 6–7; slides 19–20)

- SM1.** First error: “ $16 \div 2$ ”. Square root is not halving; it asks which number times itself is 16.  $4 \times 4 = 16$ , so  $\sqrt{16} = 4$ . (Check:  $8^2 = 64$ .)
- SM2.** First error: “It must be in the middle”.  $60 - 49 = 11$  and  $64 - 60 = 4$ , so 60 is much closer to 64 and  $\sqrt{60}$  is closer to 8.  $7.7^2 = 59.29$ ,  $7.8^2 = 60.84$ ; differences 0.71 and 0.84, so  $\sqrt{60} \approx 7.7$ . (Check:  $7.5^2 = 56.25$ , far from 60. Calculator 7.746...) Also accept “should write  $\approx$ , not  $=$ ” as a secondary comment.
- SM3.** First error: “you cannot find a root of a negative number”.  $(-2)^3 = (-2)(-2)(-2) = -8$ , so  $\sqrt[3]{-8} = -2$ . (It is square roots of negative numbers that have no real value.)

## Exit Ticket (page 8; slide 21)

- X1.** (a)  $\sqrt{49} = 7$  (b)  $\sqrt[3]{27} = 3$

**X2.**  $81 < 85 < 100$ , so  $9 < \sqrt{85} < 10$ .

**X3.** Closer to 9 ( $85 - 81 = 4$ ,  $100 - 85 = 15$ ).  $9.2^2 = 84.64$ ,  $9.3^2 = 86.49$ . Differences 0.36 and 1.49.  $\sqrt{85} \approx 9.2$ . (Calculator 9.219...) *Full credit needs a check* (a squared decimal or the reverse check  $9.2^2 = 84.64 \approx 85$ ).

**X4.** Edge =  $\sqrt[3]{8000} = 20$  cm, because  $20^3 = 20 \times 20 \times 20 = 400 \times 20 = 8000$ . *Alternative:*  $\sqrt[3]{8000} = \sqrt[3]{8} \times \sqrt[3]{1000} = 2 \times 10$ .

**Marking guide:** X1 and X2 = SC1 and SC3 (essential); X3 = SC4 (essential); X4 = transfer (SC1 + SC2).

## After-Lesson Practice

### Retrieval

**R1.** (a)  $2^4 = 2 \times 2 \times 2 \times 2 = 16$  (b)  $(-5)^2 = 25$  (c)  $0.3^2 = 0.3 \times 0.3 = 0.09$

### Easy

**Q1.** (a) 6 (b) 12 (c) 13 ( $13^2 = 169$ )

**Q2.** (a) 3 (b) 5 (c) 1

**Q3.**  $\sqrt{100} = 10$  cm.

**Q4.**  $\sqrt[3]{64} = 4$  cm.

**Q5.** (a)  $9 < 12 < 16$ : between 3 and 4. (b)  $64 < 75 < 81$ : between 8 and 9. (c)  $27 < 40 < 64$ : between 3 and 4.

### Medium

**Q6.**  $25 < 30 < 36$ , so  $5 < \sqrt{30} < 6$ ;  $30 - 25 = 5$ ,  $36 - 30 = 6$ , very slightly closer to 5 (nearly halfway).  $5.4^2 = 29.16$ ,  $5.5^2 = 30.25$ . Differences 0.84 and 0.25.  $\sqrt{30} \approx 5.5$ . (Calculator 5.477...)

**Q7.**  $64 < 70 < 125$ , so  $4 < \sqrt[3]{70} < 5$ , closer to 4.  $4.1^3 = 68.921$ ,  $4.2^3 = 74.088$ . Differences 1.079 and 4.088.  $\sqrt[3]{70} \approx 4.1$ . (Calculator 4.121...)

**Q8.**  $\sqrt{50}$  is larger.  $49 < 50$ , so  $\sqrt{50} > \sqrt{49} = 7$ .  $300 < 343$ , so  $\sqrt[3]{300} < \sqrt[3]{343} = 7$ . So  $\sqrt[3]{300} < 7 < \sqrt{50}$ .

**Q9.** (a)  $\sqrt{2500} = 50$  ( $50^2 = 2500$ ) (b)  $\sqrt{0.25} = 0.5$  ( $0.5^2 = 0.25$ ) (c)  $\sqrt[3]{8000} = 20$  ( $20^3 = 8000$ )

**Q10.** (a)  $\sqrt[3]{-64} = -4$  (b)  $\sqrt[3]{-1} = -1$   
(c)  $4^2 = 16$  and  $(-4)^2 = 16$ . Any number squared is positive (or zero), so no real number squares to  $-16$ . Therefore  $\sqrt{-16}$  has no real value.

### Hard

**Q11.** Yes, it fits. Side =  $\sqrt{12}$ .  $3.5^2 = 12.25 > 12$ , so  $\sqrt{12} < 3.5$  m. (Also  $3^2 = 9 < 12$ , so the side is between 3 and 3.5 m; calculator 3.46 m.) The rug fits, with only about 4 cm to spare.

**Q12.** First error:  $\sqrt{36 + 64} = \sqrt{36} + \sqrt{64}$ . You cannot split a square root over addition. Correct:  $\sqrt{36 + 64} = \sqrt{100} = 10$ . (Check:  $14^2 = 196 \neq 100$ .)

**Q13.** Volume =  $4 \times 4 \times 8 = 128$  cm<sup>3</sup>. Edge =  $\sqrt[3]{128}$ .  $125 < 128 < 216$ , so  $5 < \sqrt[3]{128} < 6$ , very close to 5.  $5.0^3 = 125$ ,  $5.1^3 = 132.651$ . Differences 3 and 4.651. Edge  $\approx 5.0$  cm (1 d.p.). (Calculator 5.039...) Accept "5.0" but not "5 exactly".

## Self-Test

**ST1.**  $\sqrt{121} = 11$  since  $11^2 = 121$ .

**ST2.**  $64 < 90 < 125$ , so  $4 < \sqrt[3]{90} < 5$ .

**ST3.**  $16 < 18 < 25$ : between 4 and 5, closer to 4.  $4.2^2 = 17.64$ ,  $4.3^2 = 18.49$ . Differences 0.36 and 0.49.  
 $\sqrt{18} \approx 4.2$ . (Calculator 4.242...)